Life Sciences and
Facility Management

Institute of
Computational Life Sciences

Morphological and mask operations

Image and Signal Processing

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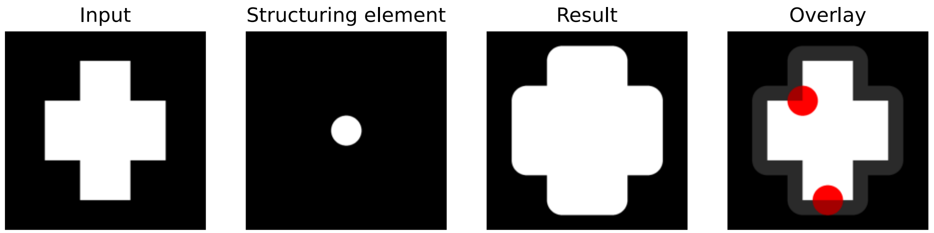
Morphological operations

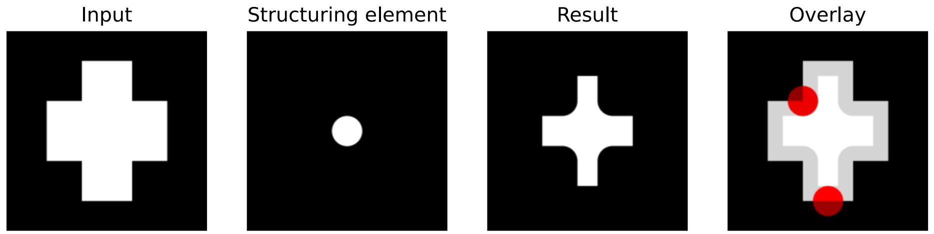
Morphological operations

- What are they?
  - Morphological operations are non-linear operators that act on images based on local spatial structure.
  - Typically applied to binary images (foreground vs. background)
- Core idea
  - Use a small pattern called a structuring element (SE)
  - Slide it over image and probe local neighborhoods
  - Modify pixels based on how the SE fits or overlaps
- Useful for post-processing segmentation masks
  - Remove noise (operation: opening)
  - Fill holes in objects (op.: closing)
  - Separate touching objects (op.: erosion)
  - Recover object size after cleaning (op.: dilation)

Dilation operation. Expands foreground regions by adding pixels to object boundaries. A structuring element slides over the image, and a pixel becomes foreground if it overlaps with any existing foreground pixel, causing objects to grow and small gaps to close.



Erosion operation. Here, in contrast, foreground regions are shrunk by removing pixels at object boundaries. A pixel remains foreground only if the structuring element fits entirely within the object, causing objects to contract and small structures to disappear.



Demo time

- Handling of segmentation masks:
  Morphological operations
- Coin segmentation / Watershed:
  Tutorial
- **Jupyter Notebooks**
  - Morphological operations
  - Mask operations